

GIN446- Web Programming



USEK — School of Engineering
Instructor: Pascal Damien

Building real, deployed web applications in the age of artificial intelligence.

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Week 4 Activity

Build Your Engineering Student Profile Form

AI Contribution Statement

The concept, design, structure, and pedagogy of this course are entirely my own. I used generative AI tools (Claude, ChatGPT, and Gemini) as an assistant to help draft, edit, and format some of these materials and to produce illustrations. Every output was reviewed, edited, and verified by me, and I remain fully responsible for the content.

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The Activity in one idea

Up to now your pages only *show* information. In this activity you'll build your first form that *collects* it. You will see the full round trip: your form sends data to the server, and the server sends back your profile page in your own design.

Part 1 — Build the form by hand

- AI level 0: No AI — you do it by hand
- Create `profile_form.html` (extending `base.html`) with the right input types, labels, grouping, and native validation, using the exact field names from the spec.

Part 2 — Style the form with AI

- AI level 2: Assisted — you direct, AI implements
- Have the AI generate a `form_style.css` for the form page, and help you debug. You audit every rule in VS Code and own the result.

The server and profile page are provided — you build the form, and everything you add later plugs into the same structure and stylesheets.

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Part 1: Build the form by hand

You'll build an HTML form (`templates/profile-form.html`) using a wide range of input types and form elements.

- `profile-form.html` extends `base.html`.
- The form gather the academic profile of Engineering Students.
- When the form is submitted (with POST), the server hands the data to a profile page (`profile.html`)
- `profile.html` extends `base.html` so the profile appears styled with your own Week 3 CSS style.

`app_week4.py` (the server) and `profile.html` (the profile page) are provided:

- Download both files from e-learning; replace the content of your `app.py` with the content of `app_week4.py` and put `profile.html` in `templates/`.

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Step 1: Build the form by hand (1/2)

Create templates/profile-form.html, extending your base template:

```
{% extends "base.html" %}
{% block title %}Engineering Student Profile{% endblock %}
{% block content %}
    <h2>Engineering Student Profile</h2>
    <p>Complete this form to provide information about your
    academic background, technical skills, internship
    experience, and Final Year Project preferences.</p>
    <form action="{{ url_for('submit_profile') }}"
        method="post">
        ...
        ...
    </form>
{% endblock %}
```

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Step 1: Build the form by hand (2/2)

Complete profile-form.html while respecting the below requirements:

- Use the EXACT names provided in the next slides. Otherwise, app.py and profile.html must be modified.
- Make sure that every control has a <label for=. . .> tied to its id.
- Use appropriate input types and native validation: required, pattern, min/max/step, maxlength.
- Use <datalist> for "Primary programming language"
- Use <select multiple> for "Engineering software used"
- Use <select> with <optgroup> for "Preferred project area" and group values by major
- Include a hidden field: <input type="hidden" name="form_version" value="2026-2027">.
- Include a required consent checkbox.
- Include a submit button and a reset button

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profile-form.html (1/5)

Senior Engineering Student – Academic Profile

Complete this form to provide information about your academic background, technical skills, internship experience, and Final Year Project preferences.

1. Personal Information

Student ID:

202312345

name="student_id"

Full Name:

First name and last name

name="fullname"

USEK Email:

student@net.usek.edu.lb

name="email"

Mobile Number:

+961 70 123456

name="phone"

Date of Birth:

dd/mm/yyyy

☐

name="birthdate"

GitHub:

https://github.com/username

name="portfolio"

GIN446 website:

Your RENDER website link

name="gin446"

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profile-form.html (2/5)

2. Academic Information

Engineering Major:

-- Select your major --

▼

name="major"

Current GPA / 100:

85.50

name="gpa"

Credits Completed:

name="credits"

Expected Graduation:

☐ Spring

☐ Summer

☐ Fall

name="graduation"

Expected Graduation Month & Year:

☐

name="graduation_month"

Have you completed an engineering internship?

☐ Yes

☐ No

name="internship_completed"

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profile-form.html (3/5)

3. Technical Background

Select all areas in which you have experience:

☐ Programming

☐ Embedded Systems

☐ Electronics

☐ Computer Networks

☐ Telecommunications

☐ Biomedical Instrumentation

☐ Artificial Intelligence / Machine Learning

☐ Signal Processing

name="skills"

Primary Programming Language:

Start typing...

name="language"

MATLAB

Simulink

LabVIEW

LTspice

Vivado

SolidWorks

Engineering Software Used:

name="software"

Hold Ctrl (Windows) or Command (Mac) to select several options.

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profile-form.html (4/5)

4. Final Year Project

Proposed Project Title:

Enter a tentative project title

name="project_title"

Preferred Project Area:

Artificial Intelligence

name="project_area"

Brief Project Description:

Describe the problem, proposed solution, and expected outcome...

name="description"

Preferred Team Size:

2

name="team_size"

Choose a color for your project dashboard:

name="project_color"

Preferred Meeting Date:

dd/mm/yyyy

name="meeting_date"

Preferred Meeting Time:

--:--

name="meeting_time"

Artificial Intelligence

Computer Engineering

Artificial Intelligence

Web Applications

Cybersecurity

Internet of Things

Electrical Engineering

Power Systems

Renewable Energy

Control Systems

Biomedical Engineering

Medical Imaging

Biosensors

Medical Devices

Telecommunications Engineering

Wireless Communications

5G / 6G Networks

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profile-form.html (5/5)

☐ I confirm that the information provided is accurate. name="confirmation"

Submit ProfileClear Form

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Step 2: Download from eLearning

Download `app_week4.py` (the server) and `profile.html` (the profile page) from e-learning.

Replace the content of `app.py` with the content of `app_week4.py`.

Put `profile.html` in `templates/`.

Explorer

Open Editors

web-programming-site

- __pycache__
- static
 - form_style.cssU
 - image.pngU
 - my_styles.cssU
- templates
 - base.htmlU
 - index.htmlM
 - internet-history-ai.htmlM
 - internet-history.htmlM
 - profile-form.htmlU
 - profile.htmlU
 - web-history-ai.htmlM
 - web-history.htmlM
- venv
- .gitignore
- app.pyM
- Profile
- README.md
- requirements.txt

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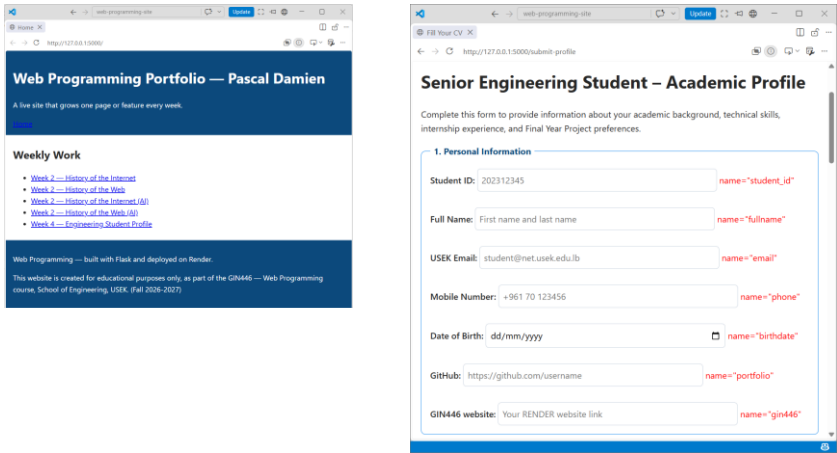
12

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Step 3 – Run and test locally

Run your app and test it (Use AI for debugging).



Part 2: Style your form with AI (1/2)

Step 1: Use AI to generate a CSS for the form (Name the file `form_style.css`).

Step 2: Make these 2 changes:

- In `base.html` (loads the site stylesheet for all pages):

```
<head>
...
< rel="stylesheet"
  href="{{ url_for('static', filename='my_styles.css') }}" />
{% block head %}{% endblock %}
</head>
```
- In `profile-form.html`

```
{% block head %}
<link rel="stylesheet"
  href="{{ url_for('static', filename='form_style.css') }}" />
{% endblock %}
```

Because `form_style.css` loads after `my_styles.css`, it can override or extend the site styles for the FORM page specifically.

Part 2: Style your form with AI (2/2)

Run your app and test it (Use AI for debugging).

Commit, push, and confirm it runs on Render.

Pascal

Computer Engineering

123456789 | pascal@netuniv.edu.au | +661 3 123456 | 2026-09-01

Academic Information

Major	Computer Engineering
GPA	90 / 100
Credits completed	99 / 148
Expected graduation	Spring — 2027-07
Internship completed	Yes

Technical Background

Areas of experience

Programming | Networks | AI | Signal Processing

Primary programming language: C

Engineering software

Matlab | Labview | Proteus

Final Year Project

Title	Artificial Intelligence
Area	AI
Preferred team size	2
Preferred meeting	2026-09-30 at 10:30
Dashboard colour	#4682B4

Description

Here is a brief description

Form version 2026-2027 — [edit and resubmit](#)

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Deliverables

The AI conversation PDF.

Your AI contribution statement.

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Acceptance checklist

☐ `profile_form.html` extends `base.html` and posts to `/submit-profile` (`method="post"`).

☐ Every field uses the exact name from the spec.

☐ Every control has a real `<label for>` tied to its `id`; the four sections and each radio group are wrapped in `<fieldset>/<legend>`.

☐ Native validation works (9-digit `student_id`, required fields, GPA 40–100, etc.).

☐ A submit button and a reset button are present.

☐ The form page has its own `form_style.css`, loaded via `{% block head %}` after the site stylesheet.

☐ Submitting shows your profile page (`profile.html`), styled with your own CSS and your chosen accent color.

☐ You can explain every AI-generated CSS rule and every fix you accepted.

☐ The page is committed, pushed, and verified live on Render, and the Profile page is linked from your home page.

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